To install Docker on an EC2 instance running Ubuntu, you can follow these general steps. I'll provide you with a basic guide for installing Docker, along with instructions for installing a specific version if needed.

1. Update the package list

First, update your existing list of packages:

bash

```
sudo apt-get update
```

2. Install necessary packages

Install packages to allow apt to use a repository over HTTPS:

bash

```
sudo apt-get install \
  ca-certificates \
  curl \
  gnupg \
  lsb-release
```

3. Add Docker's official GPG key

Add Docker's official GPG key to ensure the authenticity of the package:

bash

```
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

4. Set up the Docker repository

Use the following command to set up the Docker stable repository:

bash

```
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu \
  $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

5. Update the package list again

Update your package list to include packages from the new Docker repository:

bash

```
sudo apt-get update
```

6. Install Docker Engine

Install the latest version of Docker Engine and containerd:

bash

```
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

If you want to install a specific version of Docker, list the available versions first:

bash

```
apt-cache madison docker-ce
```

This will give you a list of available Docker versions. Then, install the specific version using:

bash

sudo apt-get install docker-ce=<VERSION_STRING> docker-ce-cli=<VERSION_STRING> containerd.io docker-buildx-plugin docker-compose-plugin

Replace `<VERSION_STRING>` with the version you want to install (e.g., `5:20.10.7~3-0~ubuntu-bionic`).

7. Verify the installation

To verify that Docker is installed correctly, run:

bash

sudo docker run hello-world

This command downloads a test image and runs it in a container. When the container runs, it prints a confirmation message.

Optional: Manage Docker as a non-root user

To run Docker commands without `sudo`, add your user to the `docker` group:

bash

sudo usermod -aG docker $USER

Log out and back in so that the group membership is re-evaluated.

--

To run any version of Docker on an EC2 instance, follow these steps. Here, I'll guide you through the process of installing a specific version of Docker and running it on an EC2 instance running Ubuntu.

Steps to Install and Run a Specific Version of Docker

1. **Connect to Your EC2 Instance:**

   First, connect to your EC2 instance using SSH.

   sh

   ssh -i /path/to/your-key.pem ubuntu@your-ec2-instance-public-dns

2. **Update Package Information:**

   Update your package list to ensure you have the latest information on the newest versions of packages and their dependencies.

   sh

   sudo apt-get update

3. **Install Required Packages:**

   Install packages to allow apt to use a repository over HTTPS.

   sh

   sudo apt-get install \

      ca-certificates \

      curl \

      gnupg \

      lsb-release

4. **Add Docker's Official GPG Key:**

   Add Docker's official GPG key to your system.

sh

sudo mkdir -p /etc/apt/keyrings

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg

5. **Set Up the Docker Repository:**

Add the Docker repository to APT sources.

sh

echo \

"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu \

$(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

6. **Update the Package List Again:**

Update your package list to include packages from the new Docker repository.

sh

sudo apt-get update

7. **List Available Docker Versions:**

List the available versions of Docker.

sh

apt-cache madison docker-ce

This command will output a list of available Docker versions.

8. **Install a Specific Version of Docker:**

Install the desired version of Docker. Replace `<VERSION_STRING>` with the version string from the previous step (e.g., `5:20.10.7~3-0~ubuntu-bionic`).

sh

```
sudo apt-get install docker-ce=<VERSION_STRING> docker-ce-cli=<VERSION_STRING> containerd.io docker-buildx-plugin docker-compose-plugin
```

9. **Verify the Installation:**

Verify that Docker is installed correctly by running a simple Docker container.

sh

```
sudo docker run hello-world
```

10. **Manage Docker as a Non-Root User (Optional):**

If you want to run Docker commands without `sudo`, add your user to the `docker` group.

sh

```
sudo usermod -aG docker $USER
```

After running this command, log out and back in so that the group membership is re-evaluated.

sh

```
exit
ssh -i /path/to/your-key.pem ubuntu@your-ec2-instance-public-dns
```

11. **Run Docker Commands:**

   Now you can run Docker commands without using `sudo`.

   sh

   docker run hello-world

Following these steps will ensure that you can install and run any version of Docker on your EC2 instance.